Multivariate Calculus Lecture Notes
Section 17.1 – Green's Theorem

I. Curl in the plane
   a. Before we talk about Green's Theorem, one of the most important theorems in all of calculus, we need to discuss more deeply the concept of *curl*.
   b. Calculating circulation for a small region
      i. Suppose we have the following rectangle:

      ii. If $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j}$, then the rates at which fluid flows along the four sides of the rectangle counterclockwise are given by:
         1. Top: Approximately $\mathbf{F}(x, y + \Delta y) \cdot (-\mathbf{i})\,\Delta x = -F_1(x, y + \Delta y)\Delta x$.
         2. Bottom: Approximately $\mathbf{F}(x, y) \cdot \mathbf{i}\,\Delta x = F_1(x, y)\Delta x$.
         3. Right: Approximately $\mathbf{F}(x + \Delta x, y) \cdot \mathbf{j}\,\Delta y = F_2(x + \Delta x, y)\Delta y$.
         4. Left: Approximately $\mathbf{F}(x, y) \cdot (-\mathbf{j})\,\Delta y = -F_2(x, y)\Delta y$.
      iii. The total rate at which fluid is flowing counterclockwise along the rectangle can be calculated via the following.
         1. Top and bottom:

$$-F_1(x, y + \Delta y)\Delta x + F_1(x, y)\Delta x = -\frac{F_1(x, y + \Delta y) - F_1(x, y)}{\Delta y}\Delta x\Delta y \approx -\frac{\partial F_1}{\partial y}\Delta x\Delta y$$

         2. Right and left:

$$F_2(x + \Delta x, y)\Delta y - F_2(x, y)\Delta y = \frac{F_2(x + \Delta x, y) - F_2(x, y)}{\Delta x}\Delta x\Delta y \approx \frac{\partial F_2}{\partial x}\Delta x\Delta y$$

         3. The total circulation is thus approximately $\left(\dfrac{\partial F_2}{\partial x} - \dfrac{\partial F_1}{\partial y}\right)\Delta x\Delta y$.
   c. Circulation density
      i. The density of the circulation (after dividing by the area of the region) is thus approximately $\dfrac{\partial F_2}{\partial x} - \dfrac{\partial F_1}{\partial y}$, and this quantity becomes exact as we let Δx and Δy go to zero.
      ii. Note: This is the circulation measured around the $\mathbf{k}$ axis.
         1. We could measure the circulation around the $\mathbf{i}$ or $\mathbf{j}$ axes as well.
         2. Of course, in the plane, we're only interested in the $\mathbf{k}$ component of circulation.
      iii. The $\mathbf{k}$ or z component of the circulation density, or *curl*, of a vector field $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j}$ at a point (x, y) is given by $\text{curl}_z\mathbf{F} = \dfrac{\partial F_2}{\partial x} - \dfrac{\partial F_1}{\partial y}$.
   d. Signs
      i. If $\text{curl}_z\mathbf{F}(x, y) > 0$ then fluid is generally flowing counterclockwise at (x, y).
      ii. If $\text{curl}_z\mathbf{F}_z(x, y) < 0$ then fluid is generally flowing clockwise at (x, y).
      iii. If $\text{curl}_z\mathbf{F}(x, y) = 0$ then fluid is not circulating or turning at (x, y).
      iv. This can be difficult to see for a specific $\mathbf{F}$.
   e. Examples

i. Suppose $\mathbf{F} = x\mathbf{i} + y\mathbf{j}$.
   1. Then $\text{curl}_z\mathbf{F} = 0 - 0 = 0$.
   2. This should make sense because the vectors in $\mathbf{F}$ are not rotating counterclockwise.
   3. In fact, a 2D vector field $\mathbf{F}$ is conservative iff $\text{curl}_z\mathbf{F} = 0$.

ii. Suppose $\mathbf{F} = y\mathbf{i} - x\mathbf{j}$.
   1. Then $\text{curl}_z\mathbf{F} = -1 - 1 = -2$.
   2. The negative sign should make sense because the vectors in $\mathbf{F}$ are rotating clockwise.

II. Green's Theorem

a. Derivation

i. Suppose we have a region D in the plane like so:

ii. Suppose we were to chop up the region into a bunch of tiny pieces:

iii. The circulation on each piece is approximately $\left(\dfrac{\partial F_2}{\partial x} - \dfrac{\partial F_1}{\partial y}\right)\Delta x \Delta y$.

iv. Suppose we add up the circulations over all the pieces and then let the sizes of the pieces get very small.

v. In the limit we would end up with

$$\iint\limits_D \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right) dA.$$

vi. However, if we compare two adjacent pieces we see that the circulations over the same boundary are negatives of each other:

vii. Thus if we were to add up the circulations over all the pieces everything would cancel out except the circulation over the boundary of D.

viii. From this point of view, in the limit we would get exactly the circulation along the boundary C: $\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C F_1 dx + F_2 dy$.

ix. This, then, is the circulation-curl form of Green's Theorem:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C F_1 dx + F_2 dy = \iint\limits_D \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right) dA = \iint\limits_D \text{curl}_z\mathbf{F}\, dA.$$

x. Integrating the z component of the curl of $\mathbf{F}$ across a region D gives the same result as calculating the counterclockwise circulation of $\mathbf{F}$ around the boundary of D.

b. Comments

i. The region D must be simply connected (i.e., no holes or crossings) with a piecewise smooth boundary curve C oriented counterclockwise.
ii. The vector field $\mathbf{F}$ must have continuous partial derivatives in an open region containing D.

III. Using Green's Theorem

a. Find the counterclockwise circulation for the field $\mathbf{F} = (x - y)\mathbf{i} + (y - x)\mathbf{j}$ along the square curve C bounded by $x = 0$, $x = 1$, $y = 0$, $y = 1$.

i. Graphically:

ii. This won't be fun as a line integral, as it requires four separate integrals.
iii. However, we can use Green's Theorem to convert to a double integral, which should be easier.
iv. We have

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C (x - y)dx + (y - x)dy = \int_0^1 \int_0^1 (-1 - (-1))\, dx\, dy = 0.$$

v. This makes sense because $\mathbf{F}$ is conservative.

b. Evaluate $\oint_C x^4\, dx + xy\, dy$, where C is the triangle (oriented counterclockwise) with vertices (0,0), (1,0), (0,1).

i. Graphically:

ii. We have $\mathbf{F} = x^4\mathbf{i} + xy\mathbf{j}$.
iii. Unlike the previous problem, $\mathbf{F}$ is not conservative.
iv. Green's Theorem, circulation-curl form
1. We take $F_1 = x^4, F_2 = xy$.
2. Then

$$\oint_C x^4\, dx + xy\, dy = \int_0^1 \int_0^{1-x} (y - 0)\, dy\, dx = \frac{1}{2}\int_0^1 (1 - x)^2\, dx = -\frac{1}{6}(1 - x)^3\Big|_0^1 = \frac{1}{6}$$

c. Find the work done by the field $\mathbf{F} = (3y - e^{\sin x})\mathbf{i} + (7x + \sqrt{y^4 + 1})\mathbf{j}$ on an object as it moves once counterclockwise around the circle of radius 3 centered at the origin.

i. Graphically:

ii. Calculating the line integral directly is going to be awful.
iii. With Green's Theorem, we have, where R is the disk of radius 3 centered at the origin,

$$W = \oint (3y - e^{\sin x})dx + (7x + \sqrt{y^4 + 1})dy = \iint_R (7 - 3)\, dA = 4\iint_R dA = 4(3^2\pi) = 36\pi.$$

d. Area with Green's Theorem

i. We know that $\iint_R dA$ gives the area of region R.

ii. Thus if we can find a vector field $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j}$ such that $\dfrac{\partial F_2}{\partial x} - \dfrac{\partial F_1}{\partial y} = 1$, we can use Green's Theorem to find the area of R by integrating around its boundary.

iii. There are an infinite number of possibilities here for F_1 and F_2.
   1. One such is $F_2 = x$, $F_1 = 0$; we can take $\mathbf{F} = x\mathbf{j}$.
   2. Another is $F_1 = -y$, $F_2 = 0$; we can take $\mathbf{F} = -y\mathbf{i}$.
   3. Yet another is $F_2 = x/2$, $F_1 = -y/2$; this yields $\mathbf{F} = -\dfrac{y}{2}\mathbf{i} + \dfrac{x}{2}\mathbf{j}$.
   4. Summary: The area enclosed by the curve C can be calculated via any of the following:
      - $\oint_C x\,dy$
      - $\oint_C -y\,dx$
      - $\frac{1}{2}\oint_C x\,dy - y\,dx$

iv. Graphically:

v. You can actually buy machines that can calculate the area of a region by traversing its boundary, using this idea; they are called *planimeters*.

vi. Example (in text): Find the area enclosed by the ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ with a line integral and the formula $\oint_C x\,dy$ for the area enclosed by C.
   1. We can parameterize the ellipse with $x = a\cos t$, $y = b\sin t$, $0 \le t \le 2\pi$.
   2. Thus $dy = b\cos t\,dt$.
   3. Thus the area of the ellipse is given by

$$\oint_C x\,dy = \int_0^{2\pi} ab\cos^2 t\,dt = ab\int_0^{2\pi} \frac{1+\cos 2t}{2}\,dt = ab\left[\frac{t}{2} + \frac{\sin 2t}{4}\right]_0^{2\pi} = \pi ab.$$

vii. See Problem #32 for a formula for the area of a generic polygon via Green's Theorem.

IV. Vector form of Green's Theorem

a. Flux across a plane curve

i. Background
   1. Intuitively, the *flux* across a plane curve is the rate at which the vector field is flowing across the curve.
   2. Graphically:

   3. A different kind of line integral can be used to calculate the flux.

4. Remember that $\oint_C \mathbf{F} \cdot \mathbf{T}\, ds$ gives the circulation; i.e.; the overall amount the vector field is going with the curve.
5. So $\oint_C \mathbf{F} \cdot \mathbf{n}\, ds$, where $\mathbf{n}$ is the outward-pointing unit normal vector, will give the flux of $\mathbf{F}$ across C.

ii. Calculating $\mathbf{n}$

1. If $\mathbf{T}$ is in the plane, the normal vector $\mathbf{n}$ is orthogonal to both $\mathbf{T}$ and $\mathbf{k}$.
2. Moreover, if the motion along the curve is counterclockwise, then $\mathbf{T} \times \mathbf{k}$ points outward, by the right-hand rule.

3. We know that if $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$, then

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|} = \frac{x'(t)\mathbf{i} + y'(t)\mathbf{j}}{\|\mathbf{r}'(t)\|}.$$

4. Thus we get, with $\mathbf{T} = T_1\mathbf{i} + T_2\mathbf{j}$,

$$\mathbf{n} = \mathbf{T} \times \mathbf{k} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ T_1 & T_2 & 0 \\ 0 & 0 & 1 \end{vmatrix} = \frac{y'(t)\mathbf{i} - x'(t)\mathbf{j}}{\|\mathbf{r}'(t)\|}.$$

5. And $\mathbf{n}$ is a unit vector, as we can see.

b. Derivation

i. We have

$$\oint_C \mathbf{F} \cdot \mathbf{n}\, ds = \int_a^b (\mathbf{F} \cdot \mathbf{n})(t)\, \|\mathbf{r}'(t)\| dt = \int_a^b \left[\frac{F_1 y'(t)}{\|\mathbf{r}'(t)\|} - \frac{F_2 x'(t)}{\|\mathbf{r}'(t)\|} \right] \|\mathbf{r}'(t)\| dt$$
$$= \int_a^b (F_1 y'(t) - F_2 x'(t)) dt = \int_C F_1 dy - F_2 dx.$$

ii. Applying the circulation-curl form of Green's Theorem, we have

$$\int_C F_1 dy - F_2 dx = \iint_D \left(\frac{\partial F_1}{\partial x} - \frac{-\partial F_2}{\partial y} \right) dA = \iint_D \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} \right) dA.$$

iii. Putting all this together, we have the flux-divergence form of Green's Theorem:

$$\oint_{\partial D} \mathbf{F} \cdot \mathbf{n}\, ds = \int_{\partial D} F_1 dy - F_2 dx = \iint_D \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} \right) dA = \iint_D \operatorname{div} \mathbf{F}\, dA,$$

where ∂D is the boundary of the region D.

iv. Just like curl is circulation density, divergence is flux density.

c. Example: Calculate the outward flux of the field $\mathbf{F}(x, y) = x\mathbf{i} + y^2\mathbf{j}$ across the square with boundaries $x = \pm 1$ and $y = \pm 1$.

i. This would require four separate line integrals.

ii. However, with Green's Theorem we have

$$\oint_{\partial D} \mathbf{F} \cdot \mathbf{n}\, ds = \int_{-1}^{1} \int_{-1}^{1} \operatorname{div} \mathbf{F}\, dx\, dy = \int_{-1}^{1} \int_{-1}^{1} (1 + 2y)\, dx\, dy = \int_{-1}^{1} x(1 + 2y) \Big|_{-1}^{1} dy$$

$$= 2\int_{-1}^{1}(1+2y)\,dy = 2\left[y+y^2\right]\Big]_{-1}^{1} = 2[1+1-(-1+1)] = 4.$$

Homework: Section 17.1 #4, 5, 6, 8, 12, 16, 19, 27, 33, 36.